



The Atlantic Blue Crab as a Cross-Ecosystem Outlier: A Framework for Wildlife Toxicology and One Health

Concepción Martínez-Gómez¹, Begoña Pérez², Lucía Viñas², Beatriz Fernández¹, Francisco Acosta-Cifuentes¹, Jessica Bargiela², Ignacio Franco¹, Antonio Zamora-López³, Claritxa Neal², Antonio A Herrero-Reyes³, Pilar Gómez-Ramírez⁴

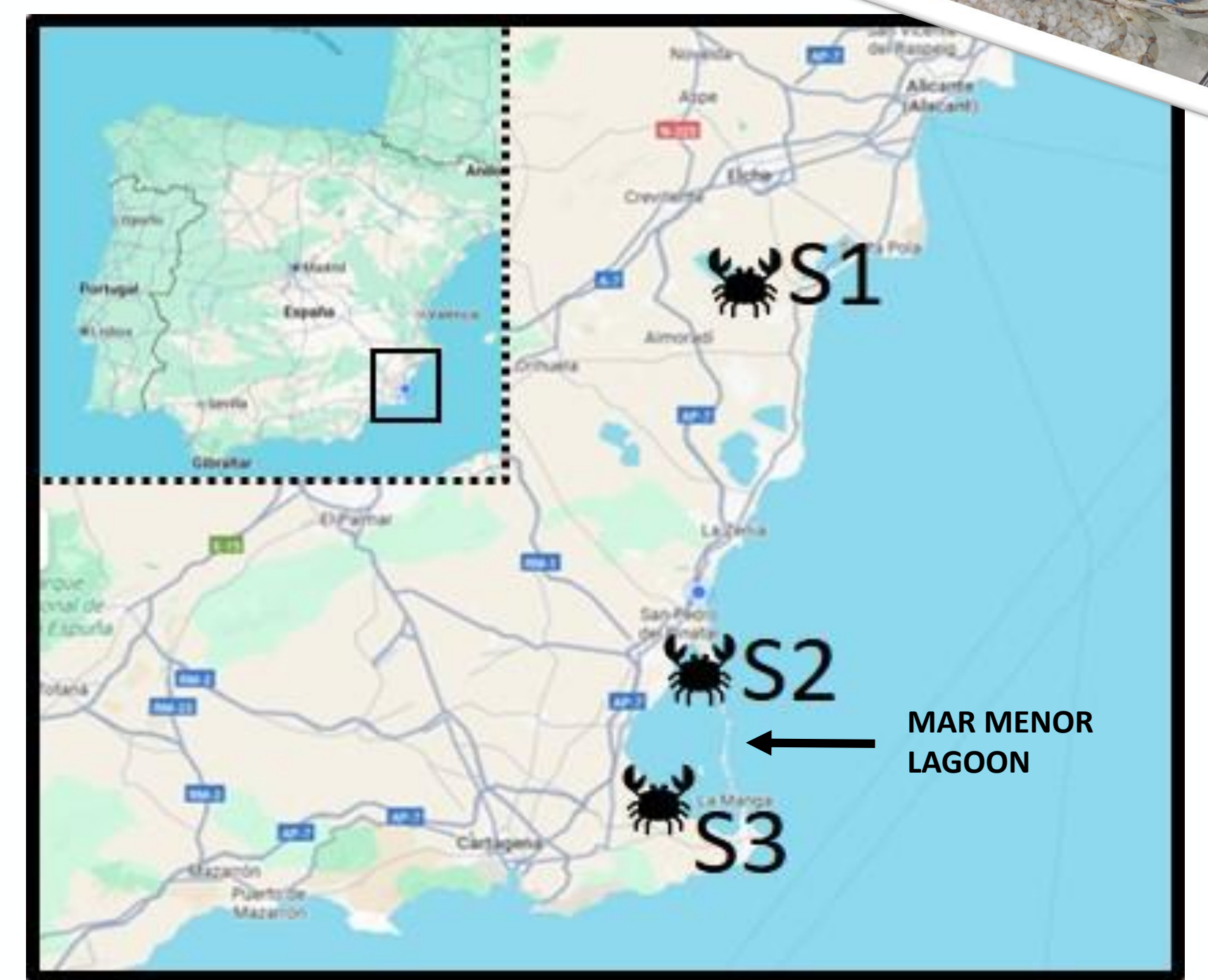
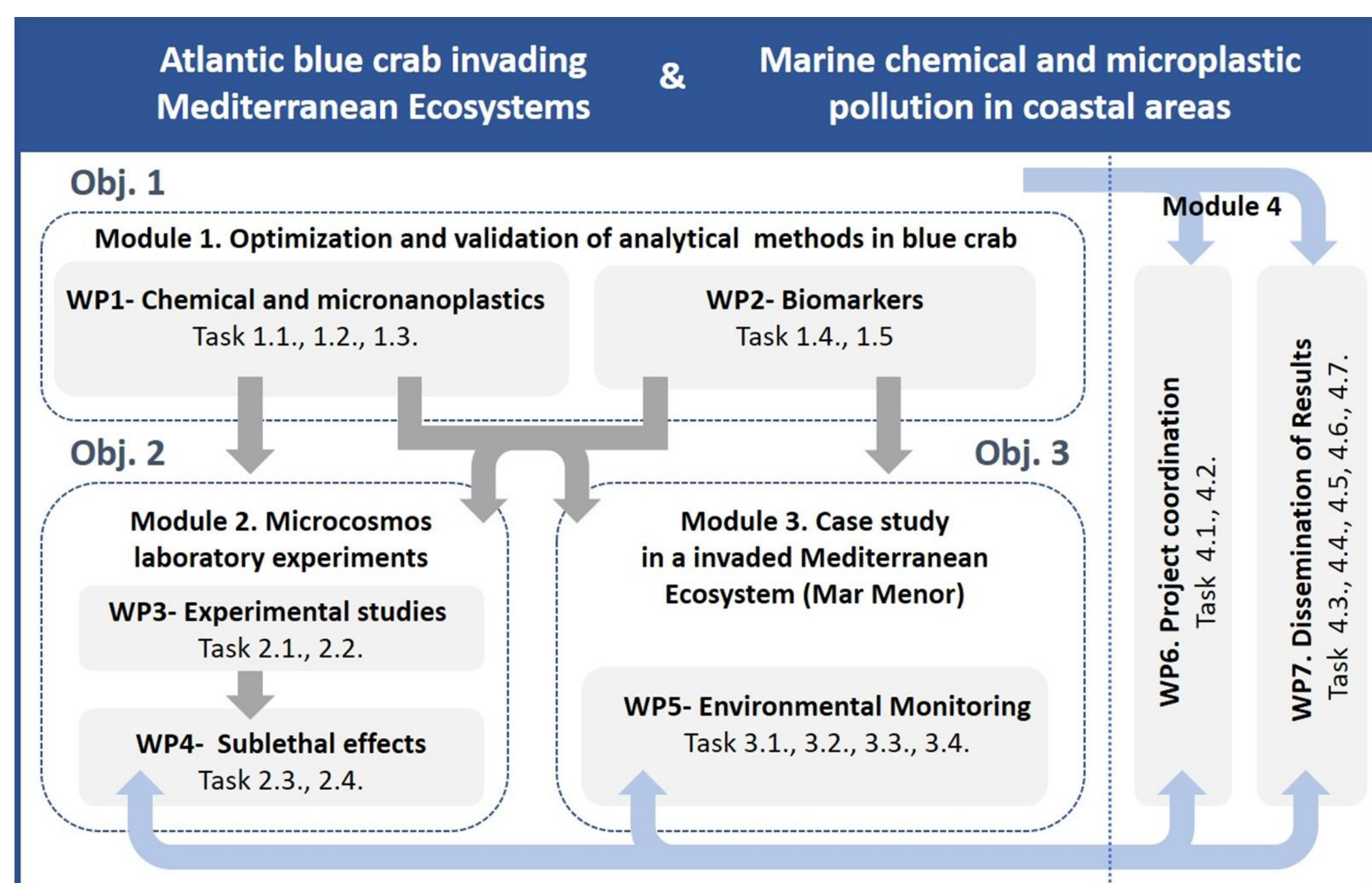
E-mail contact: concepcion.martinez@ieo.csic.es

¹ Oceanographic Centre of Murcia, Instituto Español de Oceanografía (IEO-CSIC); ² Oceanographic Centre of Vigo, Instituto Español de Oceanografía (IEO-CSIC)
³ Department of Zoology and Physical Anthropology, Faculty of Biology, University of Murcia; ⁴ Area of Toxicology, Faculty of Veterinary, Campus de Espinardo, University of Murcia

INTRODUCTION

The blue crab has features that are rarely represented in regulatory ecotoxicology, such as high metabolic plasticity (i.e. tolerance to salinity shifts) and a semi-closed circulatory system with hemocyanin-based oxygen transport, influencing toxicokinetics and a robust detoxification pathways. As omnivorous predators and scavengers, *Callinectes sapidus* experiences exposure routes that are uncommon among standard test species, with frequent interaction with contaminated sediments. The on-going CRABMEDPOL Project is based on the hypothesis that the invasive blue crab species *Callinectes sapidus* may become an excellent sentinel species for monitoring coastal pollution in invaded Mediterranean ecosystems and a good model species for marine and human toxicological research

OBJECTIVES, MATERIAL AND METHODS METHODS

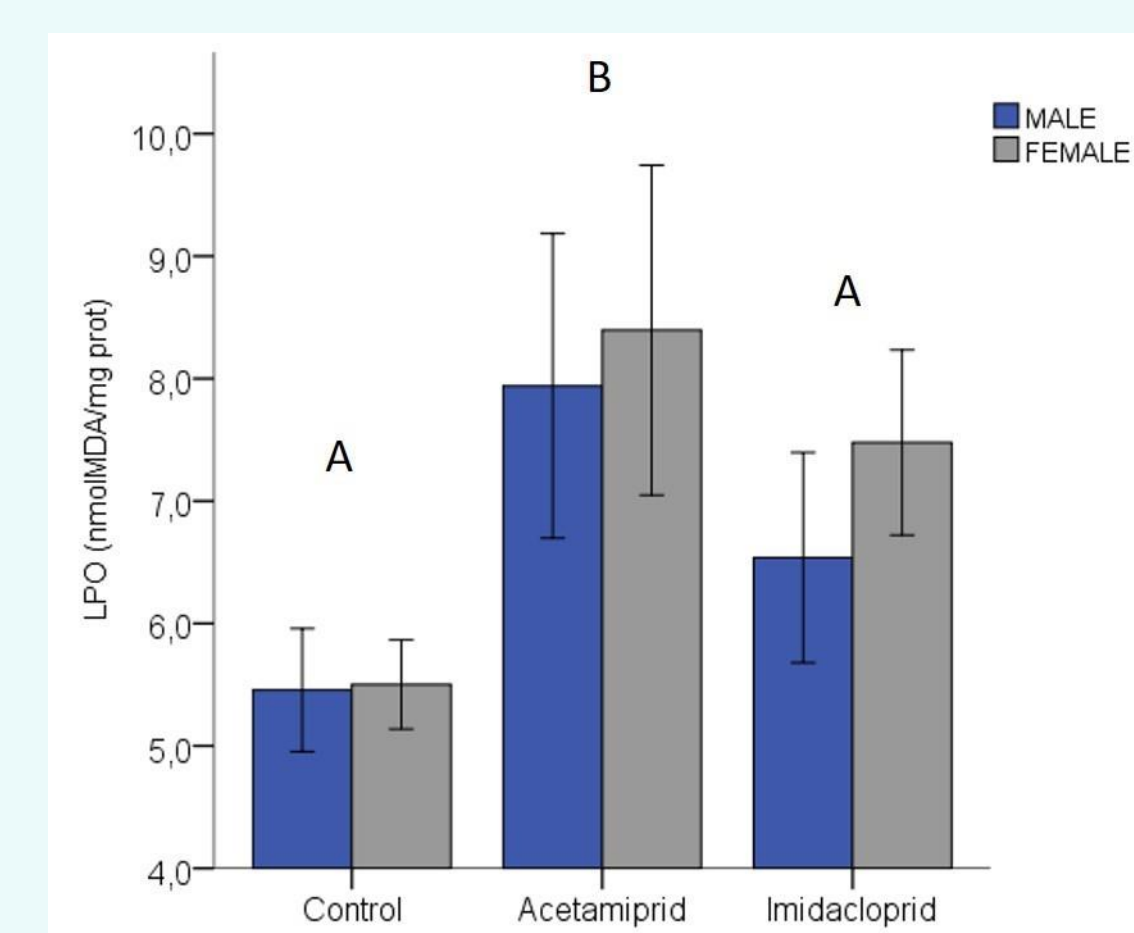


SOME PRELIMINARY AND PROMISING RESULTS

Biomarker responses in *C. sapidus* under laboratory control conditions (20°C, 42 PSU) in various tissues. (n.e =not evaluated)

Biomarker	Gills	Hepatopancreas	Muscle
AChE (nmol/min/mg)	4.02 ± 0.34	6.88 ± 0.89	24.56 ± 2.59
CAT (μmol/min/mg)	45.13 ± 3.21	141.00 ± 21.20	n.e.
GR (nmol/min/mg)	3.78 ± 0.32	4.33 ± 0.59	n.e.
LPO (nmolMDA/mg prot)	5.48 ± 0.32	4.22 ± 0.40	n.e.
SOD (U/mg)	6.50 ± 0.47	5.93 ± 0.86	n.e.
GP (nmol/min/mg)	28.91 ± 3.53	221.89 ± 15.41	n.e.
GST (nmol/min/mg)	31.18 ± 6.31	132.59 ± 7.63	n.e.

Lipid peroxidation in gills following short-term ingestion of neonicotinoid-contaminated food



- ✓ Successful maintenance of blue crab under controlled laboratory conditions for more than 90 days demonstrates its suitability for experimental studies
- ✓ The use of contaminant-spiked gelatin-based seafood is an effective method for conducting exposure studies with the blue crab.
- ✓ Bioaccumulation of neonicotinoids followed the order: hepatopancreas > muscle > hemolymph.
- ✓ Exposure to neonicotinoids may cause severe motor impairment and reduced or ceased feeding, with recovery after prolonged starvation.
- ✓ Exposure to neonicotinoids decrease Acetylcholinesterase (AChE) activity in muscle, while Lipid Peroxidation (LPO) increased significantly in the gills following exposure to Acetamiprid
- ✓ All Blue crab collected from Mar Menor contained Microplastics (MPs) in the hepatopancreas and muscle (not the gastrointestinal tract), predominantly Polypropylene fragments and fibers.
- ✓ Blue crabs from Mar Menor accumulate more Pb and Cd than in those from El Hondó Natural Park, potentially limiting for human consumption

